

DRY-RUN GUIDE | move_zero.py
Source: Python\4th_belt\move_zero.py

CORE BEHAVIOR

Moves zeros to the end while retaining the relative order of nonzero values.

REPRESENTATIVE INPUT

[0,1,0,3,12].

STEP TRACE

Collect nonzeros 1,3,12; append/fill two zeros.

EXPECTED OUTPUT / RETURN

[1,3,12,0,0]

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\4th_belt\move_zero.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Move all zeroes to the end of the array while keeping the order of non-zero elements.
002 | # Add your solution here.
003 |
004 | from typing import List
005 |
006 | def move_zeroes(arr: List[int]) -> None:
007 |     write_index = 0
008 |
009 |     for i in range(len(arr)):
010 |         if arr[i] != 0:
011 |             # Swap the non-zero element with the element at write_index
012 |             arr[i], arr[write_index] = arr[write_index], arr[i]
013 |             write_index += 1
014 |
015 | if __name__ == "__main__":
016 |     n = int(input())
017 |     arr = list(map(int, input().split()))
018 |     move_zeroes(arr)
019 |     print(" ".join(map(str, arr)))
020 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.